

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education KCSE PREDICTION SERIES 1



121/1

Mathematics

Paper 1

Time: 2½ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- Write your name, index number and date in the spaces provided at the top of this page.
- Write name, admission number and class in the spaces provided above.
- This paper contains **TWO** sections: **section I** and **section II**
- Answer **ALL** the questions in **Section I** and only **five** questions from **section II**.
- Show all the steps in your calculations, giving your answers at each stage in the spaces provided below each question.
- Marks may be given for correct working even if the answer is wrong.
- Non-programmable** silent electronic calculators and KNEC mathematical tables may be used except where stated otherwise.

FOR EXAMINER'S USE ONLY:

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

Section II

GRAND TOTAL

17	18	19	20	21	22	23	24	TOTAL

SECTION 1 (50 MARKS)

Answer all questions in this section in the spaces provided.

1. Without using a calculator or mathematical table evaluate

$$\frac{1}{3} \text{ of } (2\frac{3}{4} - 5\frac{1}{2}) \times 3\frac{6}{7} \div \frac{9}{4} \quad (3\text{mks})$$

2. Solve for x in the equation.

(3mks)

$$9^{(2x-1)} \times 3^{(2x+1)} = 243$$

3. A line P whose equation is $y = \frac{1}{3}x + 4$ is parallel to another line Q. Find the equation of line Q in the form $y = mx + c$ given that it passes through Point (3, 6) (3mks)

4. Using reciprocals, cubes and square tables, evaluate correct to 4 significant figures: **(4mks)**

$$\sqrt[3]{\frac{1}{27.38}} + 1.897^2$$

5. A point P (2, 3) is mapped onto P' (-7, 0) under an enlargement with scale factor of -2 without drawing find the centre of enlargement.

(3mks)

6. A businessman bought 100 textbooks and 80 pens for sh. 25,600. If she had bought twice as many textbooks and half as many pens she would have paid sh. 7,400 less. Find the cost of one textbook and one pen.

(3 mks)

7. The table below shows the number of faulty balls from 40 samples.

No. of faulty balls (χ)	0	1	2	3	4	5
Frequency	20	8	6	1	1	4

Calculate the mean.

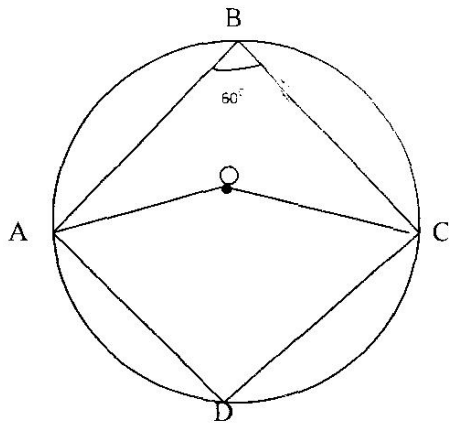
(3mks)

8. Draw a sketch and find the area in hectares of a coffee field whose measurements are entered in a field book as shown below. Take $XY = 200\text{m}$ as the baseline.

	Y	
	180	40 to Q
To R 80	140	
To S 160	100	
	40	100 to P
	X	

(4 marks)

9. In the following figure, O is the centre of the circle. Given that Angle ABC 60° , find the value of the angle ADC. **(2mks)**



10. From the top of a cliff the angle of depression of a ship when it is at A is 30° . When the ship moves 100m to point B, nearer the cliff, the angle of depression becomes 45° . Find the height of the cliff leaving your answer to 3.s.f **(4mks)**

11. During a football match, sh. 1,462,800 was realized from stadium entrance fees. If the entrance fee was sh. 80 per person, calculate how many fans paid to watch match. **(2 mks)**

12. A Kenya Bank buys and sells foreign currencies as shown

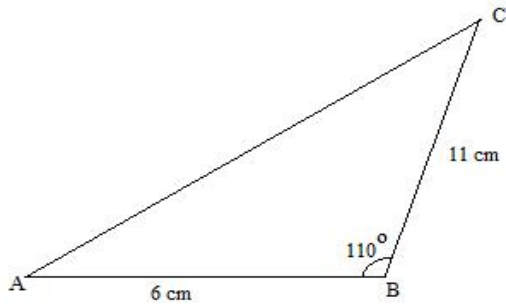
	Buying (Kshs)	Selling (Kshs)
1 Euro	84.15	84.26
100 Japanese Yen	65.37	65.45

A Japanese travelling from France to Kenya had 5000 Euros. He converted all the 5000 Euros to Kenya shillings at the bank. While in Kenya, he spent a total of Kshs. 289850 and then converted the remaining Kenya shillings to Japanese Yens at the Bank. Calculate the amount in Japanese Yen that he received.
(3mks)

13. The width of a rectangular hall of Ruiri Girls Secondary School is 16m less than its length. Calculate the length of the hall if its area is 132m^2 . Hence calculate its perimeter. (4 mks)

14. The volume of a hemisphere is 41.2cm^3 . Calculate, correct to one decimal place, the radius of the hemisphere (3mks)

- 15.** The figure below shows a triangle ABC in which $AB = 6\text{cm}$, $BC = 11\text{cm}$ and angle $ABC = 110^\circ$.
Calculate to the decimal places the length of AC. **(3mks)**



- 16.** A regular polygon has internal angle of 150° and side of length 10cm.
Find the number of sides of the polygon. **(2mks)**

Find the perimeter of the polygon. **(2mks)**

SECTION 11 (50 MARKS)

Answer any five questions from this Section.

17. A matatu left town K at 7.00a.m and travelled towards town M at an average speed of 60km/hr. A car left town M at 9.00a.m and travelled towards K at an average speed of 80km/hr. The distance between the two towns is 324km. Find:-

(a) The time each vehicle arrived at their destination

(i) Matatu

(2mks)

(ii) Car

(2mks)

b. (i) the distance the matatu had covered before the car started to move from town M to town K

(1mk)

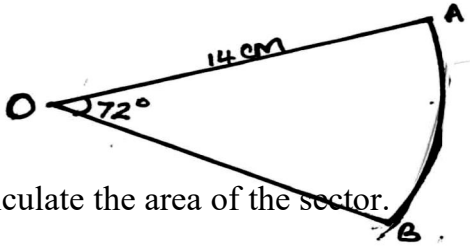
(ii) The time the two vehicles met on the way

(3mks)

(ii) How far the car was from town K when they met

(2mks)

18. The figure below shows a sector of a circle. If the radius $OA = 14\text{cm}$ and the angle $AOB = 72^\circ$.



(a) Calculate the area of the sector. (2mks)

(b) The sector is folded to form a cone. Calculate:-

(i) The radius of the cone formed. (2mks)

(ii) The volume of the solid formed. (3mks)

(c) A solid cone of same size as the one in (b) above is melted down and casted into circular washers. Each washer has an external diameter of 4cm, internal diameter of $1\frac{1}{2}\text{cm}$ and 0.3cm thick. Calculate number of washers made. (3mks)

19.a) Find the inverse of matrix $A = \begin{bmatrix} 5 & 6 \\ 7 & 9 \end{bmatrix}$ (**2mks**)

b) Okelo bought 5 physics book and six mathematics book for a total of Ksh.2440. Ali bought 7 physics book and 9 mathematics books for a total cost of ksh.3560.

i) Form a matrix equation to represent the a book **information** **(1mk)**

ii) Use matrix method to find the price of a physics book and that of a mathematic **(3mks)**

c) A school bought 36 physics books and 50 mathematics books. A discount of 5% was allowed on each Physics book whereas a discount of 8 % was allowed on each Mathematics book. Calculate the percentage discount on the cost of all the books bought. **(4mks)**

20). Forty students in a form 2 class were weighed and their masses recorded to the nearest kilogram as shown below.

45	48	56	39	47	36	45			37		46	35	43
	51	42	47	47	40	46	41	45	43	46			54
	42		51	39	42	45	44			49	50	46	39
	42	48		50		38	45		35	52	46		

a) Starting with the class 35-39 tabulate this data in a frequency table **(3mks)**

b) Find the modal class **(1mk)**

c) Calculate the mean mass of the students **(3mks)**

d) Estimate the median mass **(3mks)**

21. In triangle OAB, $OA = a$ and $OB = b$. Points P and T divide OB and AB in the ratio 2:3 and 1:3 respectively. Lines OT and AP intersect at Q.

(a) Draw the diagram to represent the above information. **(1mk)**

(b) Express OP and AP in term of a and b . **(2mks)**

(c) Express OT in terms of a and b . **(1mk)**

(d) Given further that $OQ = tOT$ and $AQ = sAP$, express OQ in two ways and hence find the values of s and t . **(6mks)**

22. A metal R is an alloy of two metals X and Y. Metal X has a mass of 70g and a density of 16g/cm^3 . Metal Y has a mass of 19g and a density of 4g/cm^3 .

(a) Calculate the density of the metal R. **(4mks)**

(b) If metal R is divided into two equal parts and each half reinforced by adding metal X to get to initial volume. Find the density of the new alloy. **(4mks)**

The two metals are mixed in a ratio of 4:1 respectively. What is the density of the alloy?
(2mks)

23. Meshach and Kelvin contributed shs. 60,000 and sh. 90,000 respectively in order to start business. They employed a manager and agreed to pay him sh. 4,500 per month from the profit made each year. They also agreed that 20% of the profit made each year would be put back into the business while the rest would be shared between them in the ratio of their initial contribution. During the first year they made a profit of sh. 365,000. Calculate:-

(a) The manager's annual salary for that year **(1mk)**

(b) The money put back into business that year. **(2mks)**

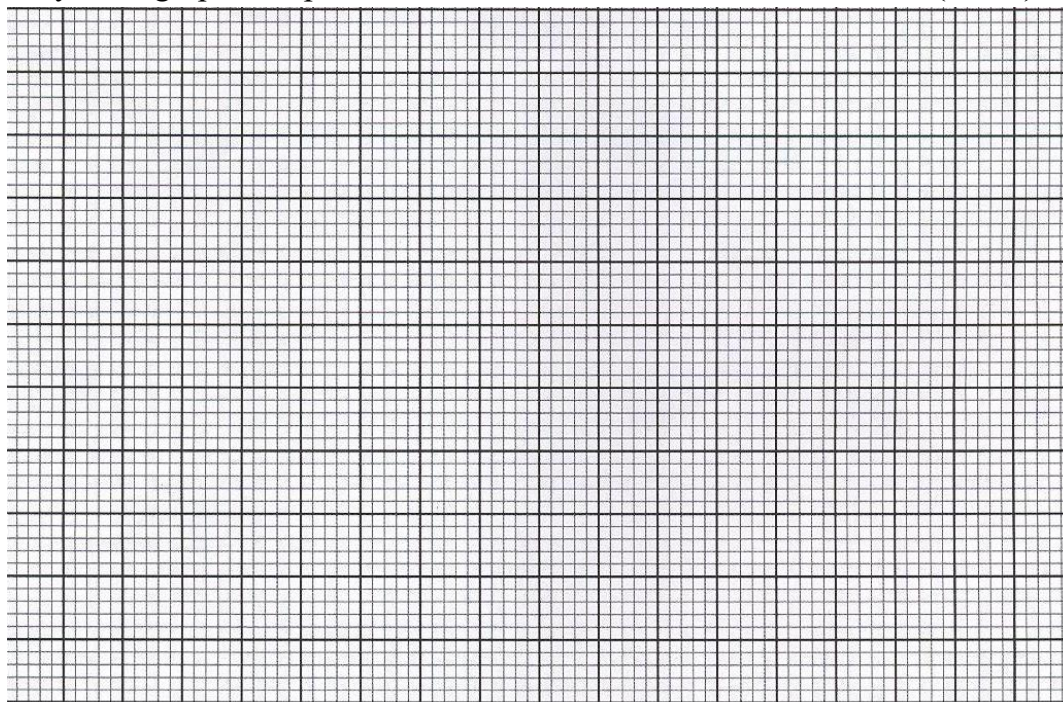
(c) The business net profit for that year. **(2mks)**

(a) How much each partner received that year. **(3mks)**

(e) The capital for the following year. **(2mks)**

24. A car starts from rest and builds up a speed of 40m/s in 1 min 40 seconds. It then travels at this speed for 5 minutes. Brakes are then applied and the car is brought to a halt in 2 minutes.

(a) Draw a velocity-time graph to represent the information above. **(3mks)**



(b) Use your graph to find

(i) The initial acceleration. **(2mks)**

(ii) The deceleration when the car is brought to a halt. **(2mks)**

(iii) The distance traveled in km. **(3mks)**